D Karthik Reddy

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PROFESSIONAL SUMMARY

Highly motivated Electronics and Communication Engineering student with proven expertise in embedded systems, AI/ML, and deep learning. Demonstrated success in developing award-winning solutions for the Smart India Hackathon, pioneering deepfake detection research, and engineering drone-based magnetic sensing systems. Adept in programming, hardware development, and real-time AI applications.

EDUCATION

VNR Vignana Jyothi Institute of Engineering and Technology — **2022 – 2026**
B.Tech in Electronics and Communication Engineering • CGPA: 9.00/10.0 — *JNTUH*

Sri Chaitanya Junior College — **2020 – 2022**
Intermediate • Percentage: 98% — *Board of Intermediate Education*

Sri Chaitanya School — **2020**
SSC • CGPA: 10.0/10.0 — *Board of Secondary Education*

TECHNICAL SKILLS

Programming: Python, C, C++, MATLAB, JavaScript, HTML, CSS, SQL, Arduino

AI/ML: TensorFlow, PyTorch, OpenCV, Scikit-Learn, Deep Learning

Embedded Systems: Raspberry Pi, Arduino, LoRa, GPS Neo-6M, Magnetometers

Tools: AutoCAD, MATLAB, NI Multisim, Xilinx Design Suite, Linux, Git

TECHNICAL PROJECTS

Deepfake Detection Using SqueezeNet — **2023**
Research Project — *Computer Vision*

- Constructed a SqueezeNet-based deepfake detection model achieving 98.21% validation accuracy.
- Trained the model on a comprehensive IEEE dataset (260K+ images) via PyTorch, optimizing hyperparameters for efficient training.
- Utilized SGDM and Adam optimizers, reducing training time to 8-11 hours per trial.
- Created a real-time detection pipeline processing 5,000 images in 14 seconds, suitable for forensic use.

Drone-Based Magnetic Sensing System — **2023**
Smart India Hackathon Winner — *Embedded Systems*

- Engineered a drone-based magnetic anomaly detection system, achieving a 40% accuracy improvement.
- Integrated a 10km-range LoRa communication system onto an 850W brushless motor drone.
- Developed real-time magnetic field mapping, achieving 95% data correlation.
- Boosted drone battery performance, extending flight time by 25%.

E-commerce Sales Prediction System — **2023**
Amazon ML Challenge — *Machine Learning*

- Built a machine learning solution (TensorFlow, NVIDIA cuDNN) improving sales prediction accuracy by 20%.
- Applied customer behavior analysis algorithms, increasing marketing campaign effectiveness by 25%.
- Established a data preprocessing pipeline to optimize model performance.

AI-based Acoustic Wave Monitoring System — **2023**
Rail Infrastructure Project — *Signal Processing*

- Fabricated an AI-driven acoustic wave monitoring system, achieving 90% accuracy in rail defect detection.
- Formulated predictive analytics for rail wear and quality assessment.

CERTIFICATIONS

Neural Networks and Deep Learning - Coursera, Andrew Ng

ACHIEVEMENTS & ACTIVITIES

Winner of Smart India Hackathon 2023 under the Ministry of Defence.

Participated in Ideathon and Convergence 2023 at VNRVJIET.

Represented school in cricket tournaments and actively engaged in badminton and cycling.